Lab 3: File System & Access Control Management

Duration: 2.5 hours

Objective:

Learn how to:

- Manage files and directories
- Create and manipulate links (hard & symbolic)
- Manage users and groups
- Understand and modify permissions
- Control ownership

Scenario :

You are a **system administrator** in a company.
You must:

- Create users and groups for departments
- Organize shared directories
- Apply secure access control policies
- Ensure proper collaboration between teams

Part 1 – Files & Directories

1. Create the following directory structure :

```
/home/lab/company
├── hr
├── it
└── finance
```

2. Inside each directory:

   - Create 2 empty files
   - Create one file with content

3. Show detailed info about the created structure

Part 2 – Links

1. In /home/lab/company/hr, create a file employees.txt
2. Create:

A hard link → employees_hard.txt

A symbolic link → employees_soft.txt

3. Delete the original file and observe behavior

4. What happens to each link after deletion? Why?

Part 3 – Users & Groups

1. Create the three following groups: **hr_team, it_team, finance_team**

2. Create the following users:

   - `alice` (HR employee)
   - `bob` (IT employee)
   - `charlie` (Finance employee)

3. Assign each user to their respective group

4. Create an additional shared group**: management**

5. Add `alice` and `bob` to `management`

6. What is the difference between primary and secondary groups?

Part 4 – Permissions & ownership

1. Apply permissions:

   - Only HR group can access `/hr`
   - Only IT group can access `/it`
   - Finance directory is **read-only**

2. Set:

   - Owner: respective department user
   - Group: respective department group

3. Test access:

- Login as different users
- Try accessing other directories

4. Give management group read access to all directories

Part 5 – Access Control

A confidential file must:

- Be readable only by alice
- Be editable by management
- Be invisible to others

Create it and apply the correct:

- Ownership
- Group
- Permissions